

Design Pattern 1: Singleton Pattern

Exercise 1: Implementing the Singleton Pattern

Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

Steps:

Create a New Java Project:

Create a new Java project named SingletonPatternExample.

Define a Singleton Class:

Create a class named Logger that has a private static instance of itself.

Ensure the constructor of Logger is private.

Provide a public static method to get the instance of the Logger class.

Implement the Singleton Pattern:

Write code to ensure that the Logger class follows the Singleton design pattern.

Test the Singleton Implementation:

Create a test class to verify that only one instance of Logger is created and used across the application.